

ECEN 142L: Communications and Networking Lab

Lab 2: 5G Technologies

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Objective

The goal of this lab is to learn about the work going into 5G development and gain a better understanding of the challenges involved in realizing a real-life communication system. The lab is completed individually, though discussion with classmates is allowed.

Lab Overview/Instructions

For this lab, you will explore part of the 5G standard and select one or two sections that seem most interesting. After reviewing those sections, you will research the topic further so you can better understand the ideas behind the standard and how they relate to real-world communication systems.

The final report should be between 1.5 and 2 pages (excluding Overview and Objective), does not need in-text citations, and should include any useful outside resources at the end of the document. The final submission must be uploaded as a PDF on Camino.

Selected Topic(s) from the 5G Standard

- Network Slicing
- Multiple Access Technologies

Network Slicing Summary

If a standard has network slicing, then a network operator can split a network into various individual customized networks that serve different requirements on performance, functionality, or accessibility.

Network slicing must support providing connectivity to home and roaming users in the same slice, support IMS (Integrated Management Systems) as a part of the slice, and support IMS independently. Slicing must also allow the operator to create, modify, and delete the slice at all times and properly coordinate across slice networks.

Key benefits of network slicing involve allowing enterprises to tailor slices to their needs, optimize resources, and accelerate time to market delivery. Because network slicing allows customization within the broader network infrastructure, vendors can tailor slices to meet diverse application needs and serve more industries overall. The customization also allows for more optimized utilization of resources, which allows the slices to be more efficient and thus deliver services only where needed (less financial waste). Lastly, due to the customization of these slices, time-to-market for these network based services are accelerated as all requirements needed for a given service can be implemented with more ease.

Some key use-cases of network slicing involve multi-enterprise campus connectivity

services, remote broadcasting, mobile cloud and gaming. Across multi-enterprise campuses, there are various networking devices and IoT (Internet of Things) devices that all need to be connected for completely different reasons with different permissions. Network slicing allows this to happen neatly all through one large network and individual slices. The same goes for remote broadcasting, which requires high bandwidth networks. With slicing, you can define the remote broadcasting slice as a high bandwidth slice without delegating the whole network to the same bandwidth requirements, avoiding bogging down the whole network. The same principle applies for mobile cloud gaming. Gaming does not require as high of bandwidth, but it does require low latency. With slicing, you can ensure the slice for mobile gaming is extremely low latency while allowing other slices to not have the same rigorous latency expectations, which overall helps the network be more efficient.

Multiple Access Technologies Summary

Multiple access technologies are advanced methods that allow thousands of devices connected to a network to occupy the same RF (radio frequency) spectrum without interference whilst delivering high bandwidth and low latency. These methods include 3GPP (3rd Generation Partnership Project) technologies including one or more NR (New Radio) and E-ULTRA (Evolved Universal Terrestrial Radio Access), as well as non-3GPP access technologies. The 5G system selects and steers to the most appropriate 3GPP or non-3GPP access technology for a service.

It is required, to properly adhere to this standard, that the 5G system shall enable the UE (user equipment) to select, manage, and efficiently provision services over the 3GPP or non-3GPP access. This means it must work in a static and dynamic networking state, and optimally distribute traffic across the network. The system must also support seamless handover across NR and E-ULTRA, satellite access, and fixed broadband access.

A key benefit of multiple access technologies is flexibility. 5G can support 3GPP and non-3GPP access technologies, so the networks are not limited to one method of connectivity, and instead can choose the most appropriate technology depending on the service, device, and network conditions. This makes the network more efficient and adaptable, which helps improve performance. Traffic can be distributed in a way that better matches the needs of the application, whether that means lower latency, better coverage, or more reliable service, and thus makes the whole system more optimal and cost-effective. 5G also supports a much wider range of devices and environments than older generations of wireless systems due to these various multiple access technologies. This ability for backwards compatibility allows system-wide adoption for corporations and state infrastructure as their legacy systems can remain unaffected by these changes.

Multiple access technologies are particularly useful in situations where devices need to move across different network environments or where one type of access may work better than another. For example, a user driving along a highway between different coverage areas can stay connected more smoothly due to the system managing handovers between access technologies efficiently. In other cases where wide area connectivity is crucial, cellular access

may be strong in one place while satellite or fixed broadband access may be more useful in another, and the ability to choose which demonstrates real life flexibility. This is also very useful for IoT and other specialized devices, since different devices may have very different communication needs. Overall, multiple access technologies help 5G provide broader coverage, smoother mobility, and better service in practice.

References

3GPP. (2019, December). TS 22.261 V17.1.0: Service requirements for the 5G system; Stage 1 (Release 17).

Ericsson. (n.d.). Network slicing. Ericsson. Retrieved April 17, 2026.

3GPP. (n.d.). 5G System Overview. Retrieved April 17, 2026.